Biweekly Assessment BW01 — Weeks 1–2

Sessions covered: S01–10
Topics: Computer foundations, terminal, JDK 17, `javac` / `java`, `main`, variables, primitives, operators, `Scanner`, casting, TechMart canteen calculator

Timebox: 45–60 minutes (homework or Friday of the even week).
Rules: Closed notes unless your trainer allows open notes. No copying.
Coding constraint: Linear `main` only — no `if` / `while` / `for` / extra methods (those start Week 3+).

A. Multiple Choice (12)

Circle the best option.

1. A computer mainly follows which pattern?
   a) Input → Process → Output
   b) Output → Input only
   c) Process without input
   d) Hardware only
2. Which extension is a Java **source** file?
   a) `.java`
   b) `.exe`
   c) `.class` only
   d) `.txt`
3. In the terminal, which command lists files (Unix/macOS)?
   a) show
   b) list
   c) dir
   d) ls
4. `javac HelloJava.java` produces:
   a) Only a PDF
   b) `HelloJava.class` bytecode
   c) A MySQL table
   d) A Git commit
5. Which is a valid **variable** name in Java?
   a) `2marks`
   b) `studentName`
   c) `student-name`
   d) `class`
6. `5 / 2` when both sides are `int` is:
   a) `2.5`
   b) `2`
   c) `3`
   d) An error always
7. `Integer.parseInt("78.5")` typically:
   a) Returns `78`
   b) Returns `78.5`

c) Throws `NumberFormatException`
d) Returns `null`

8. GST 5% on subtotal uses which idea?
   a) `subtotal + 5`
   b) `subtotal * 0.05` added to subtotal (or `* 1.05`)
   c) `subtotal / 5`
   d) `subtotal - 0.05`
9. `System.out.println` is used to:
   a) Show output on the screen
   b) Create folders
   c) Install the JDK
   d) Read keyboard input
10. `Scanner.nextLine()` gives you a:
    a) `String`
    b) `boolean` always
    c) `int` always
    d) `double` always
11. VS Code / IntelliJ Community is mainly a:
    a) Web browser only
    b) Database server
    c) Code editor / IDE
    d) Operating system
12. The JVM starts your program at:
    a) Any method named `start`
    b) `public static void main(String[] args)`
    c) The first comment
    d) `System.out` only

B. Coding (2)

1. Ask for a **TechMart** canteen item **price** (`double`) and **qty** (`int`). Compute total with **5% GST** and print `Total: ₹…` with 2 decimals. Linear `main` only (no `if` / extra methods).

```
// your code
```

2. Read three subject marks with `Double.parseDouble` and print their average with 2 decimals (`printf`). No loops / extra methods.

```
// your code
```

C. Short answer (1)

In 3–4 sentences, explain the difference between an **algorithm** and a **program**, with one India campus example.

(Write 3–6 sentences.)